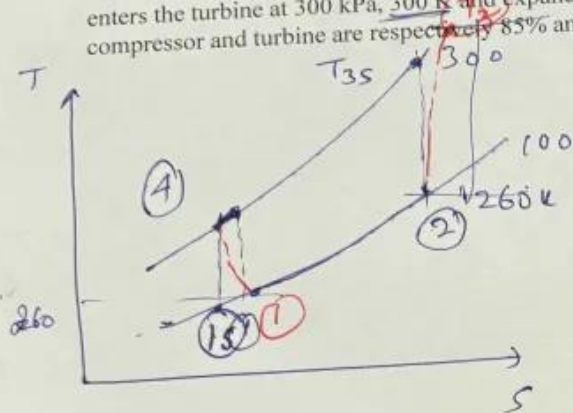


Air enters the compressor of a Brayton refrigeration cycle at 100 kPa, 260 K and is compressed adiabatically to 300 kPa. Air enters the turbine at 300 kPa, 300 K and expands adiabatically to 100 kPa. For the cycle if the isentropic efficiencies of the compressor and turbine are respectively 85% and 88%, determine the air flow required for a cooling load of 200 kW.



$$\frac{T_{3s}}{T_2} = \left(\frac{P_3}{P_2}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow \frac{T_{3s}}{260} = \left(\frac{300}{100}\right)^{\frac{0.4}{1.4}} \Rightarrow T_{3s} = 356 \text{ K}$$

$$\eta_c = \frac{h_{3s} - h_2}{h_3 - h_2} = \frac{T_{3s} - T_2}{T_3 - T_2} \Rightarrow 0.85 = \frac{356 - 260}{T_3 - 260}$$

$$\Rightarrow \boxed{T_3 = 373 \text{ K}}$$

$$\left(\frac{T_4}{T_{1s}}\right) = \left(\frac{P_4}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow T_4 = 219 \text{ K}$$

$$\eta_T = \frac{h_4 - h_1}{h_4 - h_{1s}} = \frac{T_4 - T_1}{T_4 - T_{1s}} \Rightarrow 0.88 = \frac{300 - T_1}{300 - 219}$$

$$\Rightarrow \boxed{T_1 = 229 \text{ K}}$$

$$w_c = c_p(T_3 - T_2)$$

$$= (1.005)(373 - 260) = 113.6$$

$$w_T = c_p(T_4 - T_1)$$

$$= (1.005)(300 - 229) = 71.6$$

$$w = w_c - w_T = 42 \text{ kJ/kg}$$

$$q_c = h_2 - h_1 = (1.005)(260 - 229) = 31.2$$

$$\text{COP} = \frac{q_c}{w} = \frac{31.2}{42} = 0.74$$

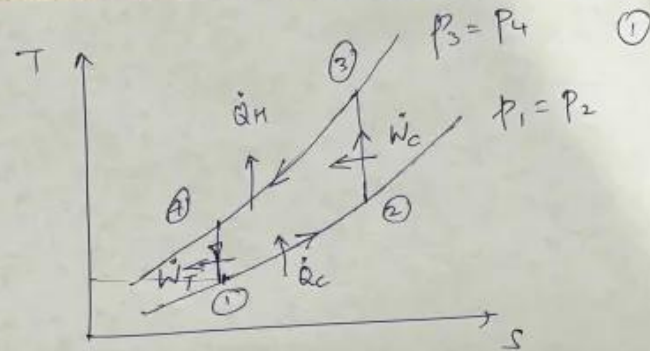
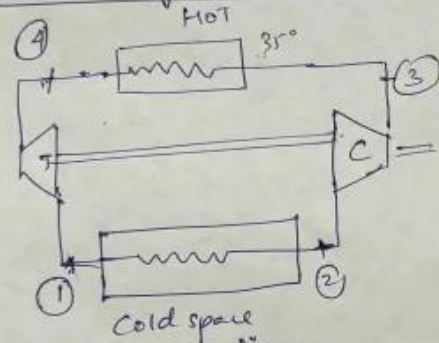
Air flow

$$\dot{Q}_c = \dot{m} q_c$$

$$(200) = (\dot{m})(31.2)$$

$$\dot{m} = 6.4 \text{ kg/s}$$

Air refrigeration



- 1-2: Constant pressure heat addition
- 2-3: Isentropic compression
- 3-4: Const. press. heat rejection
- 4-1: Isentropic expansion

$$h_1 - h_2 + q_c = 0 \Rightarrow q_c = h_2 - h_1$$

$$h_2 - h_3 - (-w_c) = 0 \Rightarrow w_c = h_3 - h_2$$

$$h_3 - h_4 - q_h = 0 \Rightarrow q_h = h_3 - h_4$$

$$h_4 - h_1 - w_T = 0 \Rightarrow w_T = h_4 - h_1$$

$$\text{Net work} = w_c - w_T = (h_3 - h_2) - (h_4 - h_1)$$

$$\dot{m}(h_1 + \frac{v_1^2}{2} + gz_1) - \dot{m}(h_2 + \frac{v_2^2}{2} + gz_2) + \dot{Q} - \dot{W} = 0$$

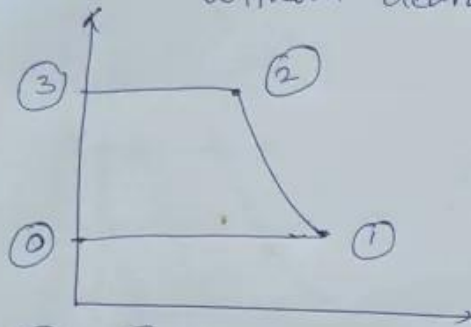
$$(h_1 + \frac{v_1^2}{2} + gz_1) - (h_2 + \frac{v_2^2}{2} + gz_2) + q - w = 0$$

$$q = \frac{\dot{Q}}{\dot{m}} \quad w = \frac{\dot{W}}{\dot{m}}$$

$$\text{COP}_{\text{refrig}} = \frac{q_c}{w_{\text{net}}} \quad \text{COP}_{\text{heat}} = \frac{q_h}{w_{\text{net}}}$$

If $\dot{Q}_c$ is the cooling load (in kW) then $\dot{Q}_c = \dot{m} q_c$, $\dot{W} = \dot{m} w$

without clearance

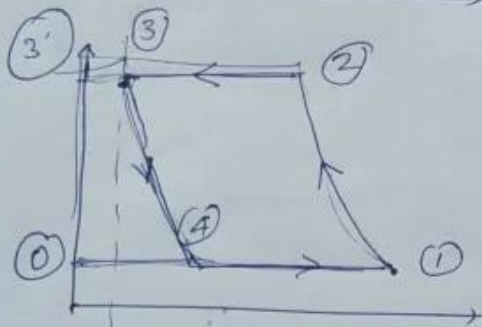


$$W = mRT \frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad (1)$$

m = mass of air taken in

T = temp of inlet air

$r \triangleq P_2/P_1$ = pressure ratio.

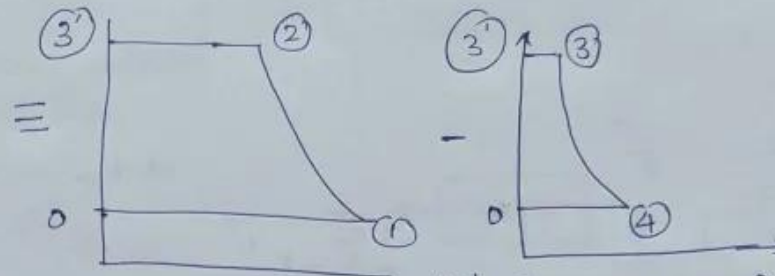


V_c clearance volume
 V_s swept volume

$$V_c + V_s = V_1$$

$$\eta_{vol} = \frac{V_1 - V_4}{V_s} = 1 - K(r^{\frac{1}{n}} - 1)$$

$$K = \frac{V_c}{V_s} \text{ clearance ratio}$$



$$= m_1 RT \frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right] - m_4 RT \frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$= (m_1 - m_4) \frac{RT}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

Total work done

$$W = W_{LP} + W_{HP}$$

$$W = \frac{n}{n-1} mRT_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right] + \frac{n}{n-1} mRT_5 \left[\left(\frac{P_6}{P_5} \right)^{\frac{n-1}{n}} - 1 \right] \quad (3)$$

If intercooling is perfect $T_1 = T_5$

$$P_5 = P_2$$

$$W = \frac{n}{n-1} mRT_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} + \left(\frac{P_6}{P_2} \right)^{\frac{n-1}{n}} - 2 \right]$$

What value of intermediate pressure P_2 gives minimum work?

$$\text{Let } k = \frac{n-1}{n}$$

$$\text{Let } W_1 = \left(\frac{P_2}{P_1} \right)^k + \left(\frac{P_6}{P_2} \right)^k - 2$$

$$\frac{dW}{dP_2} = 0$$

(to find P_2 that minimizes W)

$$\Rightarrow \frac{dW_1}{dP_2} = 0 \Rightarrow$$

$$\frac{1}{P_1} k P_2^{k-1} + P_6^k P_2^{-k-1} (-k) = 0$$

$$\frac{1}{P_1} k P_2^{k-1} = P_6^k P_2^{-k-1}$$

$$P_2^{2k} = P_1^k P_6^k$$

$$(P_2^{2k})^{\frac{1}{k}} = (P_1^k P_6^k)^{\frac{1}{k}}$$

$$P_2^2 = P_1 P_6$$

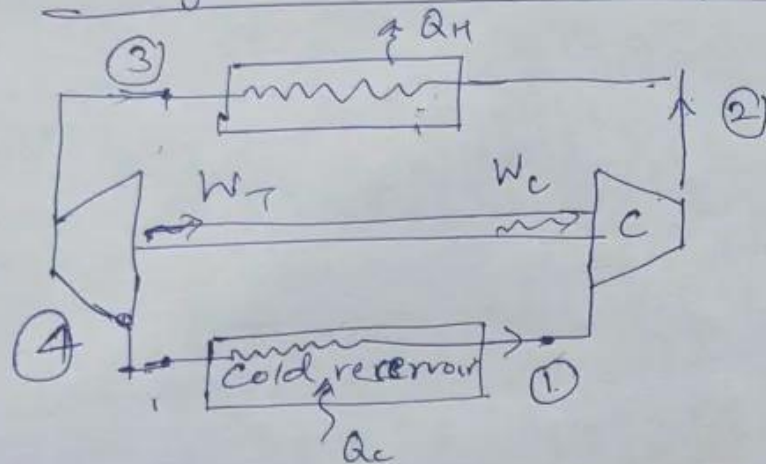
$$\Rightarrow \boxed{P_2 = \sqrt{P_1 P_6}}$$

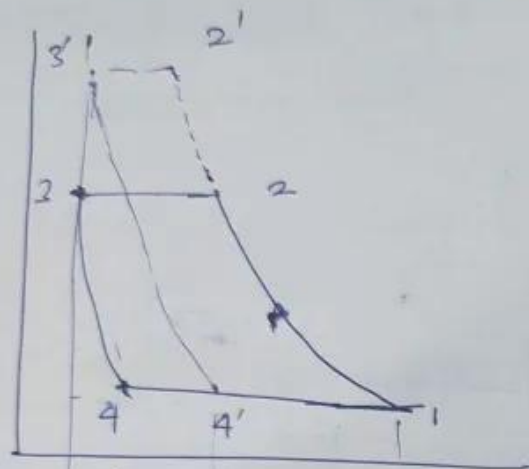
geometric mean of P_1 & P_6

Refrigeration cycle

Air cycle: Reversed Brayton cycle

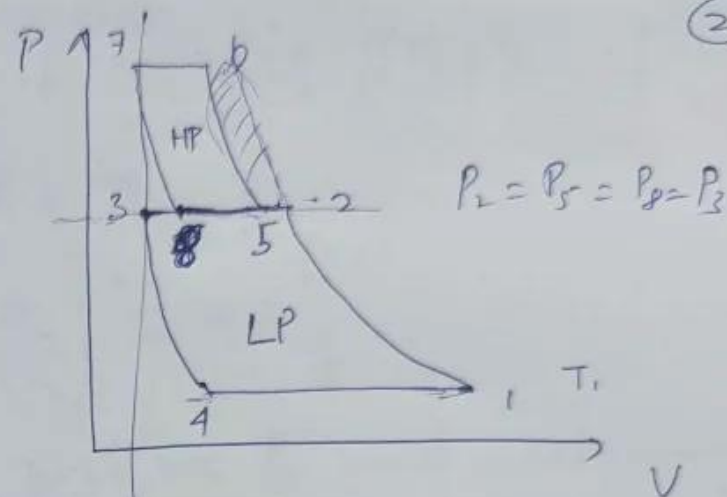
④



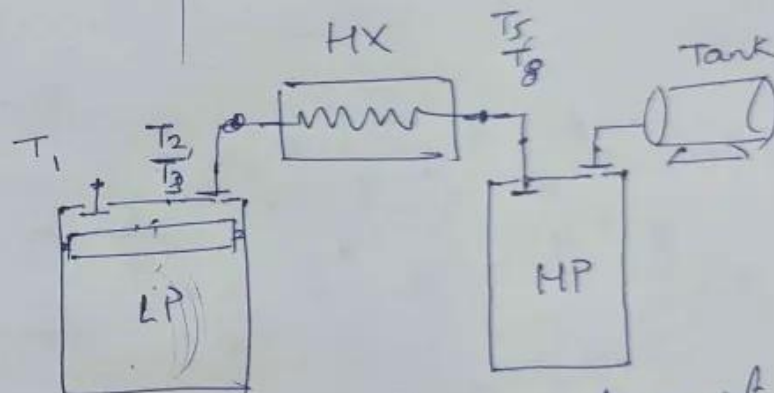


Volume intake
for P_2/P_1

Volume intake
for P_2'/P_1



$$P_2 = P_5 = P_8 = P_3$$



Perfect intercooling = temperature at
exit of intercooler = temperature at
inlet of LP cylinder
i.e. $T_5, T_8 = T_1, T_4$